

0.1 常用 Taylor 级数

1. $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^k}{k!} + \cdots.$
2. $\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + \frac{(-1)^k}{k+1} x^{k+1} + \cdots, x \in (-1, 1].$
3. $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots + \frac{(-1)^k}{(2k+1)!} x^{2k+1} + \cdots.$
4. $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots + \frac{(-1)^k}{(2k)!} x^{2k} + \cdots.$
5. $\tan x = \sum_{n=0}^{\infty} \frac{(-1)^{n-1} 2^{2n} (2^{2n-1} - 1) B_{2n}}{(2n)!} x^{2n-1} = 2 \sum_{n=0}^{\infty} \frac{(4^n - 1)(2n)!}{(2n+1)!} x^{2n+1} = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \frac{62}{2835}x^9 + \frac{1382}{155925}x^{11} + o(x^{11}), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$
6. $\cot x = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n} B_{2n}}{(2n)!} x^{2n-1} = \frac{1}{x} - \frac{1}{3}x + \frac{1}{45}x^3 - \frac{2}{945}x^5 - \cdots, x \in (0, \pi).$
7. $\sec x = \sum_{n=0}^{\infty} \frac{(-1)^n E_{2n}}{(2n)!} x^{2n} = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \frac{61}{720}x^6 + \frac{277}{8064}x^8 + \frac{50521}{3628800}x^{10} + o(x^{11}), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), x \in (0, \pi).$
8. $\csc x = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2 (2^{2n-1} - 1) B_{2n}}{(2n)!} x^{2n-1} = \frac{1}{x} + \frac{1}{6}x + \frac{7}{360}x^3 + \frac{31}{15120}x^5 + \cdots, x \in (0, \pi).$
9. $\arcsin x = \sum_{n=0}^{\infty} \frac{(2n)!}{4^n (2n+1)!} x^{2n+1} = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \frac{5}{112}x^7 + \frac{35}{1152}x^9 + \frac{63}{2816}x^{11} + o(x^{11}), x \in (-1, 1).$
10. $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \cdots + \frac{(-1)^k}{2k+1} x^{2k+1} + \cdots, x \in (-1, 1).$
11. $\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = x + \frac{1}{6}x^3 + \frac{1}{120}x^5 + \cdots + \frac{x^{2n+1}}{(2n+1)!} + \cdots.$
12. $\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots + \frac{x^{2k}}{(2k)!} + \cdots.$
13. $\tanh x = \sum_{n=0}^{\infty} \frac{4^n (4^n - 1) B_{2n}}{(2n)!} x^{2n-1} = x - \frac{1}{3}x^3 + \frac{2}{15}x^5 - \frac{17}{315}x^7 + \frac{62}{2835}x^9 - \frac{1382}{155925}x^{11} + o(x^{11}), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$
14. $\operatorname{sech} x = \sum_{n=0}^{\infty} \frac{E_{2n} x^{2n}}{(2n)!} = 1 - \frac{1}{2}x^2 + \frac{5}{24}x^4 - \frac{61}{720}x^6 + \frac{277}{8064}x^8 - \frac{50521}{3628800}x^{10} + o(x^{11}), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$
15. $\operatorname{arsinh} x = \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{4^n (2n+1)!} x^{2n+1} = x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \frac{5}{112}x^7 + \frac{35}{1152}x^9 - \frac{63}{2816}x^{11} + o(x^{11}), x \in (-1, 1).$
16. $\operatorname{artanh} x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} = x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \frac{1}{7}x^7 + \frac{1}{9}x^9 + \frac{1}{11}x^{11} + o(x^{11}), x \in (-1, 1).$
17. $e^{\sin x} = 1 + x + \frac{1}{2}x^2 - \frac{1}{8}x^4 - \frac{1}{15}x^5 - \frac{1}{240}x^6 + \frac{1}{90}x^7 + \frac{31}{5760}x^8 - \frac{1}{5670}x^9 - \frac{2951}{3628800}x^{10} + o(x^{10}).$
18. $e^{\tan x} = 1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{3}{8}x^4 + \frac{37}{120}x^5 + \frac{59}{240}x^6 + \frac{137}{720}x^7 + \frac{871}{5760}x^8 + \frac{41641}{3628800}x^9 + o(x^9).$
19. $e^{\arcsin x} = 1 + x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{5}{24}x^4 + \frac{1}{6}x^5 + \frac{17}{144}x^6 + \frac{13}{126}x^7 + \frac{629}{8064}x^8 + \frac{325}{4536}x^9 + o(x^9).$

20. $e^{\arctan x} = 1 + x + \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{7}{24}x^4 + \frac{1}{12}x^5 + \frac{29}{144}x^6 - \frac{1}{1008}x^7 - \frac{1249}{8064}x^8 - \frac{1163}{72576}x^9 + o(x^9).$
21. $\tan(\tan x) = x + \frac{2}{3}x^3 + \frac{3}{5}x^5 + \frac{181}{315}x^7 + \frac{59}{105}x^9 + \frac{3455}{6237}x^{11} + o(x^{11}).$
22. $\sin(\sin x) = x - \frac{1}{3}x^3 + \frac{1}{10}x^5 - \frac{8}{315}x^7 + \frac{13}{2830}x^9 - \frac{47}{49896}x^{11} + o(x^{11}).$
23. $\tan(\sin x) = x + \frac{1}{6}x^3 - \frac{1}{40}x^5 - \frac{107}{5040}x^7 - \frac{73}{24192}x^9 + \frac{41897}{39916800}x^{11} + o(x^{11}).$
24. $\sin(\tan x) = x + \frac{1}{6}x^3 + \frac{1}{40}x^5 - \frac{55}{846}x^7 - \frac{143}{3456}x^9 - \frac{968167}{39916800}x^{11} + o(x^{11}).$
25. $(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!} x^k + \cdots, x \in (-1, 1).$
26. $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1.$
27. $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - x^3 + \cdots, \quad |x| < 1.$
28. $(1+x)^{\frac{1}{x}} = e - \frac{e}{2}x + \frac{11e}{24}x^2 - \frac{7e}{16}x^3 + \frac{2447e}{5760}x^4 - \frac{959e}{2304}x^5 + \frac{281343e}{580608}x^6 - \frac{67223e}{168885}x^7 + o(x^7).$
29. $(1+x^2)^{\frac{1}{x}} = 1 + x + \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{11}{24}x^4 + \frac{11}{120}x^5 + \frac{271}{720}x^6 + \frac{53}{2320}x^7 - \frac{4069}{13410}x^8 + o(x^8).$
30. $(1+\sin x)^{\frac{1}{x}} = e - \frac{e}{2}x + \frac{7e}{24}x^2 - \frac{3e}{16}x^3 + \frac{139e}{560}x^4 - \frac{899e}{11520}x^5 + \frac{29811e}{580608}x^6 - \frac{180617e}{580608}x^7 + o(x^7).$